

BharatMart Enterprise Retail Intelligence Project Documentation

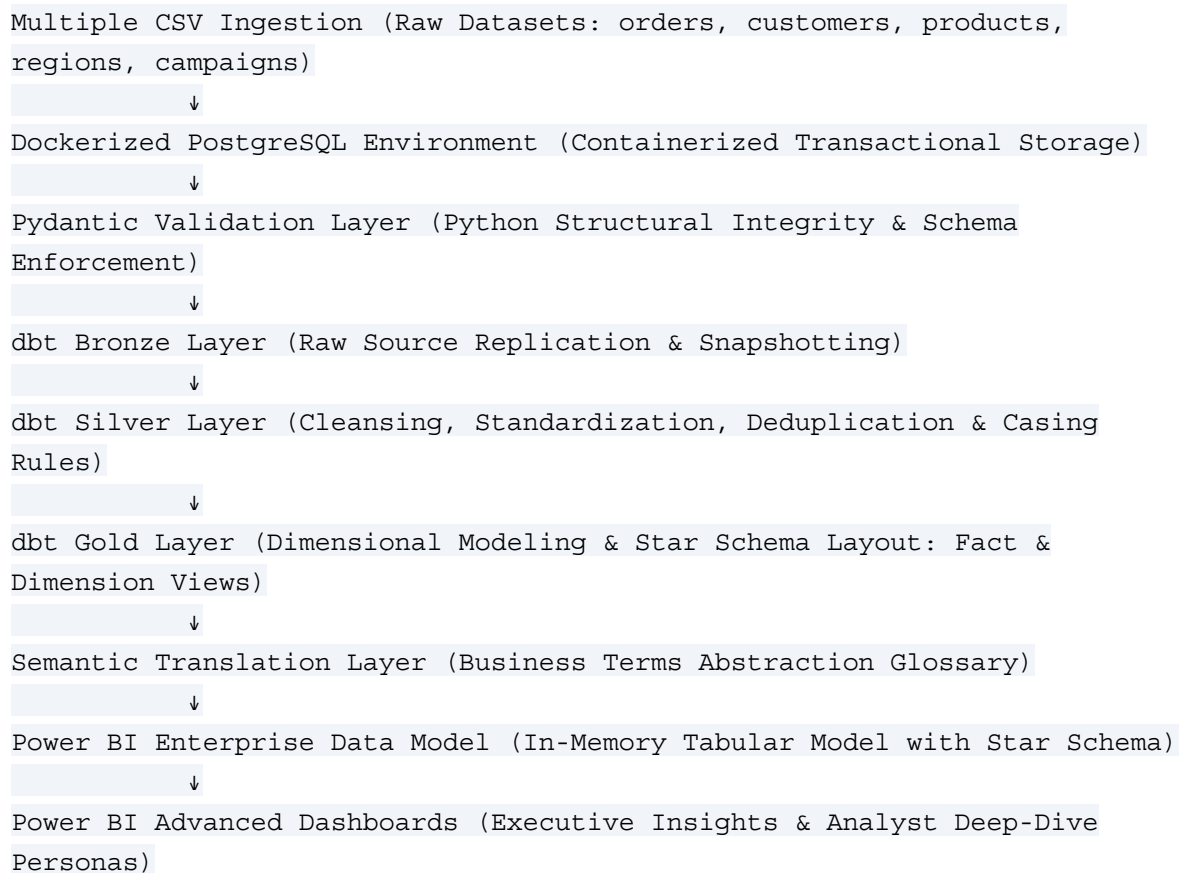
Generated on: 13 June 2026

Project Version: V6 (The Ultimate Master Blueprint)

Project Scope: End-to-End Analytics, Medallion Architecture, Advanced DAX Enterprise Models, and Multi-Persona BI Solution

1. End-to-End Project Architecture & Folder Structure

The BharatMart project architecture models a highly realistic, modern Indian enterprise retail platform. It simulates the absolute flow of operational business data through structured transformations, data governance rules, and semantic translation layers, leading to production-grade data products and persona-driven dashboards.



Production-Grade Folder Structure

- data/ — Contains all operational source flat files extracted from transactional logs.
- postgres/ — Contains the DDL initialization configurations and scripts.
- ingestion/ — Python execution scripts performing custom parsing, logging, and schema constraints.
- dbt_project/ — Orchestration code for structural views, indexing execution, and data quality tests.
- semantic_layer/ — Logical metadata schema dictionaries.
- powerbi/ — Production-ready .pbix data models containing relationships and DAX scripts.
- notebooks/ — Analytical notebooks used for exploratory analysis.

2. Intentional Data Quality Flaws & dbt Actions

Issue Identified	Automated dbt Rectification Action
Null City Values	Replaces empty strings and null records with a standardized 'Unknown' identifier.
Duplicate Orders	Implements strict window functions (ROW_NUMBER()) to deduplicate source rows via unique order_id keys.
Negative Quantities	Captures operational adjustments and cross-references with the returned_flag.
Mixed Date Formats	Parses varying patterns into uniform ISO-standard DATE data types.
Uppercase States	Applies clean text-transform format standards (Proper Upper/Lower Casing text constraints across all regional attributes).

Issue Identified	Automated dbt Rectification Action
Orphaned Foreign Keys	Assigns a default 'Unknown Customer/Product' surrogate key to maintain referential integrity.

3. Seasonal Business Logic & Storytelling Framework

- **Summer Seasonality:** Triggers a sharp volume and product revenue breakout across cooling appliances in hot regional zones.
- **Winter Essentials:** Prompts swift revenue shifts toward household heating appliances inside Northern regions.
- **Diwali & Festive Dhamaka:** Generates extreme compounding transaction clusters across high-end consumer Electronics and Fashion segments.
- **Wedding Bonanza:** Drives an upward shift in Average Order Value (AOV) tracking high-tier luxury goods.
- **Monsoon Impacts:** Triggers prominent logistical supply chain delays and drops general demand for offline apparel segments.
- **IPL (Indian Premier League) Windows:** Creates clear demand peaks for modern TVs, home entertainment electronics, and associated consumer goods.

4. Extended Semantic Layer Dictionary

Table Target	Technical Column	Business Semantic Name
orders_fact	quantity	Units Sold
orders_fact	unit_price	Base Selling Price
orders_fact	delivery_days	Fulfillment Lead Time (Days)

Table Target	Technical Column	Business Semantic Name
orders_fact	order_date	Transaction Date
orders_fact	returned_flag	Return Indicator
orders_fact	customer_rating	Customer Satisfaction Score
customers_dim	cust_acquisition_channel	Source of Acquisition
products_dim	category / sub_category	Product Line / Segment
campaigns_dim	campaign_name	Marketing Initiative
regions_dim	warehouse_region	Logistics Hub

5. Master Metric Governance & Ultimate DAX Model

```
-- 1. Net Revenue Measure Calculation
Net_Revenue =
CALCULATE(
    SUMX('orders_fact', 'orders_fact'[Units Sold] * 'orders_fact'[Base Selling
Price] - 'orders_fact'[discount]),
    'orders_fact'[order_status] <> "Cancelled"
)

-- 2. Gross Profit Margin %
Gross_Profit_Margin =
DIVIDE(
    [Net_Revenue] - SUMX('orders_fact', 'orders_fact'[Units Sold] *
RELATED('products_dim'[unit_cost])),
    [Net_Revenue],
    0
)
```

```

-- 3. Discount Penetration %
Discount_Penetration_Rate =
DIVIDE(
    SUM('orders_fact'[discount]),
    SUMX('orders_fact', 'orders_fact'[Units Sold] * 'orders_fact'[Base Selling
Price])),
    0
)

-- 4. Average Order Value (AOV) Measure
Average_Order_Value =
DIVIDE(
    [Net_Revenue],
    CALCULATE(DISTINCTCOUNT('orders_fact'[order_id]), 'orders_fact'[order_status]
<> "Returned"),
    0
)

-- 5. Delivery SLA Success Rate Measure
Delivery_SLA_Percentage =
DIVIDE(
    CALCULATE(COUNT('orders_fact'[order_id]), 'orders_fact'[Fulfillment Lead Time
(Days)] <= 3, 'orders_fact'[order_status] = "Delivered"),
    CALCULATE(COUNT('orders_fact'[order_id]), 'orders_fact'[order_status] =
"Delivered"),
    0
)

-- 6. Repeat Customer Rate
Repeat_Customer_Rate =
VAR CustomersWithMultipleOrders =
    CALCULATE(
        DISTINCTCOUNT('orders_fact'[customer_id]),
        FILTER(
            VALUES('orders_fact'[customer_id]),
            CALCULATE(DISTINCTCOUNT('orders_fact'[order_id])) > 1
        )
    )
RETURN
    DIVIDE(CustomersWithMultipleOrders, DISTINCTCOUNT('orders_fact'[customer_id]),
0)

-- 7. Active Customer Count (90-Day Rolling Window)
Active_Customers_90D =
CALCULATE(
    DISTINCTCOUNT('orders_fact'[customer_id]),
    FILTER(

```

```

        ALL('date_dim'[Date]),
        'date_dim'[Date] >= MAX('date_dim'[Date]) - 90 &&
        'date_dim'[Date] <= MAX('date_dim'[Date])
    )
)

-- 8. Year-to-Date (YTD) Revenue
YTD_Revenue =
TOTALYTD([Net_Revenue], 'date_dim'[Date])

-- 9. YoY Revenue Growth % (Time Intelligence)
YoY_Revenue_Growth =
VAR RevenueLastYear = CALCULATE([Net_Revenue],
SAMEPERIODLASTYEAR('date_dim'[Date]))
RETURN
    DIVIDE([Net_Revenue] - RevenueLastYear, RevenueLastYear, 0)

-- 10. Campaign Return On Investment (ROI)
Campaign_ROI =
DIVIDE(
    [Net_Revenue] - SUM('orders_fact'[shipping_cost]),
    DISTINCTCOUNT('orders_fact'[campaign_id]) * 50000,
    0
)

```

6. Multi-Persona Dashboard Wireframe & Interaction Rules

A. The Executive Persona View (Designed for C-Suite/VPs)

- **Primary KPIs Injected:** Total Net Revenue, YoY Revenue Growth, Gross Profit Margin, Repeat Customer Rate.
- **Visual Elements Used:** Clean metric cards, continuous line trends mapping Year-on-Year performance, horizontal category breakdown.
- **Interaction Level:** Filter parameters restricted to high-tier fields (Fiscal Quarter, Macro Zone, Product Category).

B. The Data Analyst Deep-Dive View (Operational/Power User)

- **Primary KPIs Injected:** Detailed cross-category margins, Discount Penetration, localized delivery distribution tracking.

- **Visual Elements Used:** Interconnected Matrix visual components, conditional-formatted scatter plots, localized heatmaps.
- **Interaction Level:** Full slicing control, advanced text searches, and automated page Drill-Through mechanics.

7. User Acceptance Testing (UAT) Verification Plan

```
-- UAT Case 1 Validation Script: Auditing Net Revenue Aggregations against BI
Cards
SELECT
    SUM(quantity * unit_price - discount) AS raw_validated_net_revenue
FROM public.orders_fact
WHERE order_status <> 'Cancelled';

-- UAT Case 2 Validation Script: Verifying Seasonal Category Performance
Constraints
SELECT
    p.category,
    SUM(o.quantity) AS total_seasonal_units
FROM public.orders_fact o
JOIN public.products_dim p ON o.product_id = p.product_id
WHERE o.order_date BETWEEN '2025-05-01' AND '2025-07-31'
GROUP BY p.category;

-- UAT Case 3 Validation Script: Validating Deduplication Effectiveness via Window
Integrity
SELECT
    order_id,
    COUNT(*)
FROM public.orders_fact
GROUP BY order_id
HAVING COUNT(*) > 1; -- Expected result set: 0 rows returned after dbt Execution.
```

8. Enterprise Data Governance & Security Controls (Row-Level Security)

Role Identifier	Allowed Data Visibility Scope	Governing DAX Expression Filter
Executive / C-Suite	Full corporate visibility across all metrics and all geographical regions.	No restriction filtering applied.
Regional Manager (North Hub)	Restricted strictly to performance markers within North Hub operations.	'regions_dim'[warehouse_region] = "North Hub"
Marketing Campaign Auditor	Can analyze operational campaign associations but has zero access to private customer data attributes.	Field-level metadata restriction (Omits Personal Identifiable Information columns).